MILITARY SPECIFICATION

IDOXURIDINE AND POLYVINYL ALCOHOL OPHTHALMIC SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Idoxuridine and Polyvinyl Alcohol Ophthalmic Solution (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals (See 6.3).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Phila. PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a sterile ophthalmic solution of Idoxuridine 0.1 percent and Polyvinyl Alcohol 1.4 percent and shall be suitable for topical use. Preservative and its concentration shall be acceptable to the Food and Drug Administration (FDA).

3.1.1 The ophthalmic solution shall be essentially free from foreign particles and shall be in accordance with USP requirements for ophthalmic solutions.

3.2 Assay of the finished solution.

3.2.1 Idoxuridine. Shall assay to contain not less than 0.100 percent and not more than 0.115 percent of idoxuridine at time of delivery, and not less than 0.095 percent and not more than 0.115 percent of Idoxuridine for the remainder of the expiration dating period, when assayed as specified in 4.5.1.

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3.2.2 Polyvinyl alcohol. Shall assay to contain not less than 1.20 percent and not more than 1.60 percent of polyvinyl alcohol, when assayed as specified in 4.5.2.

3.3 Chromatographic homogeneity of idoxuridine (thin layer chromatography in the finished solution). Shall show only a single spot at the same R_f value as the Idoxuridine Reference Standard, and no secondary degradation spots when determined as described in 4.5.3.

3.4 pH. The pH of the solution shall be between 5.00 and 6.00, at time of delivery, and between 4.00 and 6.00 for the remainder of the expiration dating period when determined potentiometrically, at 25°C., using the USP method.

3.5 Sterility. The ophthalmic solution shall be sterile when tested in accordance with the USP Sterility Test Method for "Liquids and Suspensions."

3.6 Ingredients. All ingredients entering into the manufacture of the solution shall be of USP or NF quality, or if not included in either of these compendia, they shall be of a high quality and purity and suitable for use in the solution.

3.7 Expiration dating period. The expiration dating period shall be 36 months.

3.7.1 Delivery. Not less than 33 months of the expiration dating period shall remain at the time of delivery to the Government.

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3.8 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

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4.3.1 For end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. End item testing

Characteristic	Requirement paragraph	Inspection level 1/	Minimum Sample
Assay:			
Idoxuridine	3.2.1	S-2	5
Polyvinyl alcohol	3.2.2	S-2	5
Chromotographic homogeneity	3.3	S-2	5
pH	3.4	S-2	5
Sterility	3.5	-	-

Composite

1/Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects

Categories	Defects 1/
Major	
101	Solution not essentially free from foreign matter.
102	Solution has crystal formation.
103	Solution not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents and foreign matter.
202	Label not free from defacing marks and tears.

1/Inspection is not restricted to the classified possible defects listed above.

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4.5 Tests.

4.5.1 Assay for Idoxuridine in the finished solution.

Standard preparation. Accurately weigh about 50 mg of Idoxuridine U.S.P. Reference Standard into a 250-ml volumetric flask, dilute to volume with 0.1N sulfuric acid, and mix well until dissolved. Pipet 25.0 ml of this solution into a 250-ml volumetric flask, dilute to volume with 0.1N sulfuric acid, and mix well.

Sample preparation. Pipet 5.0 ml of solution into a 250-ml volumetric flask, dilute to volume with 0.1N sulfuric acid, and mix well.

Procedure. Determine the maximum absorbance of each solution at 288 ± 1 nm, versus the solvent, using 1-cm cells in a suitable spectrophotometer.

Calculation:

$$\frac{Au}{As} \times \frac{\text{mg of std}}{250} \times \frac{25}{250} \times \frac{250}{\text{ml of sample}} \times \frac{100}{1000} = \text{percent of idoxuridine}$$

Where:

A_u - absorbance of sample.
A_s - absorbance of std.

4.5.2 Assay for polyvinyl alcohol in the finished solution. Shall be determined as follows:

Standard preparation. Accurately weigh about 70 mg of polyvinyl alcohol (20-90 Grade) Reference Standard into a 100-ml volumetric flask, add about 80 ml of hot (85° C.) purified water, mix until dissolved, cool to room temperature, dilute to volume, and mix well.

Sample Preparation. Pipet 5.0 ml of sample into a 100-ml volumetric flask, bring to volume with purified water, and mix well.

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Procedure. Pipet 1.0 ml of the Standard Preparation into a 50-ml volumetric flask, and 1.0 ml of the Sample Preparation into a second 50-ml flask. To each of the Blank, add the following in the order given: 25 ml of purified water; 15.0 ml of 3.2 percent boric acid solution; 0.50 ml of a solution in 1.27 percent iodine and 2.5 percent potassium iodide. Dilute each flask to volume with purified water, mix well, and allow to stand for 5 minutes. Concomitantly determine the absorbances of the Standard and Sample solutions at 660 nm, in a suitable spectrophotometer, using the Blank solution to set the instrument.

Calculation:

$$\frac{Au}{As} \times \frac{\text{mg of std}}{100} \times \frac{100}{\text{ml sample}} \times \frac{100}{1000} = \text{percent of polyvinyl alcohol}$$

Where:

Au - absorbance of sample
As - absorbance of std.

4.5.3 Chromatographic homogeneity in the finished solution.

Coat a plate, about 20 by 10 cm, or alternately 20 by 20 cm in size, with a layer of silica gel approximately 250 microns thick containing suitable phosphors. As an alternate, Eastman Chromogram Sheets type 6060, or an equivalent, shall be used.

Standard preparation. Prepare a solution containing 2.0 mg of Idoxuridine U.S.P. Reference Standard in 2 ml of 1.4 percent polyvinyl alcohol reference standard in a test tube. Add 3 ml of 4 percent boric acid solution, shake vigorously and filter.

Sample preparation. Pipet 2 ml of ophthalmic solution into a second test tube, add 3 ml of 4 percent boric acid solution, shake vigorously, and filter.

Procedure. Using pipets or syringes, apply six 10 ul portions of each filtrate, air drying after each application, to form two spots about 5 cm apart, and 2 cm from the edge of the plate. Allow the spots to dry completely and then place the plate vertically in a suitable developing tank containing

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Procedure (cont'd)

4:1 chloroform-isopropanol. Line the inside walls of the tank with filter paper saturated with the solvent. The level of the solvent shall be about 1 cm from the bottom of the tank. Place the lid on the tank and allow the solvent front to move up the plate at least 15 cm, then remove the plate, and allow it to dry at room temperature for a few minutes. Examine the plate using a shortwave (254 nm) ultraviolet light source. Circle the spots.

4.6 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE IV. Examination of preparation for delivery1/

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

1/ The above examination is only applicable to the unit, intermediate and exterior containers.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be Level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 15 ml, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available ophthalmic dropper container with an integral dropper tip, considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue shall be packaged as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred forty-four (144) units of issue shall be packaged as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the national stock number.

NOTES: (1) The national stock number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the national stock number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings shall be shown as specified in MIL-STD-129. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F.)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking shall be shown as specified in MIL-STD-129. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F.)".

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-112-9325	IDOXURIDINE AND POLYVINYL ALCOHOL OPHTHALMIC SOLUTION, 15 ml

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1642

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-I-37326A IDOXURIDINE AND POLYVINYL ALCOHOL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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